

```

#include<iostream.h>
#include<conio.h>
#define max 4
class multiple_stack
{
    int a[max],top1,top2,x;
public:
    multiple_stack()
    { top1=-1;
      top2=max;
    }
    void push1();
    void push2();
    void pop1();
    void pop2();
    void show1();
    void show2();
};

void multiple_stack::push1()
{
    if(top1+1==top2)
    {
        cout<<"stack overflow";
    }
    else
    {
        cout<<"enter value";
        cin>>x;
        top1++;
        a[top1]=x;
    }
};

void multiple_stack::push2()
{
    if(top2==top1+1)
    {
        cout<<"stack overflow";
    }
    else
    {
        cout<<"enter value";
        cin>>x;
        top2--;
        a[top2]=x;
    }
};

void multiple_stack::pop1()
{
    if(top1==0)
    {
        cout<<"stack overflow";
    }
    else
    {
        x=a[top1];
        top1--;
    }
};

```

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void multiple_stack::pop2()
{
    if(top2==max)
    {
        cout<<"underflow stack";
    }
    else
    {
        x=a[top2];
        top2++;
        cout<<x<<"removed element";
    }
}

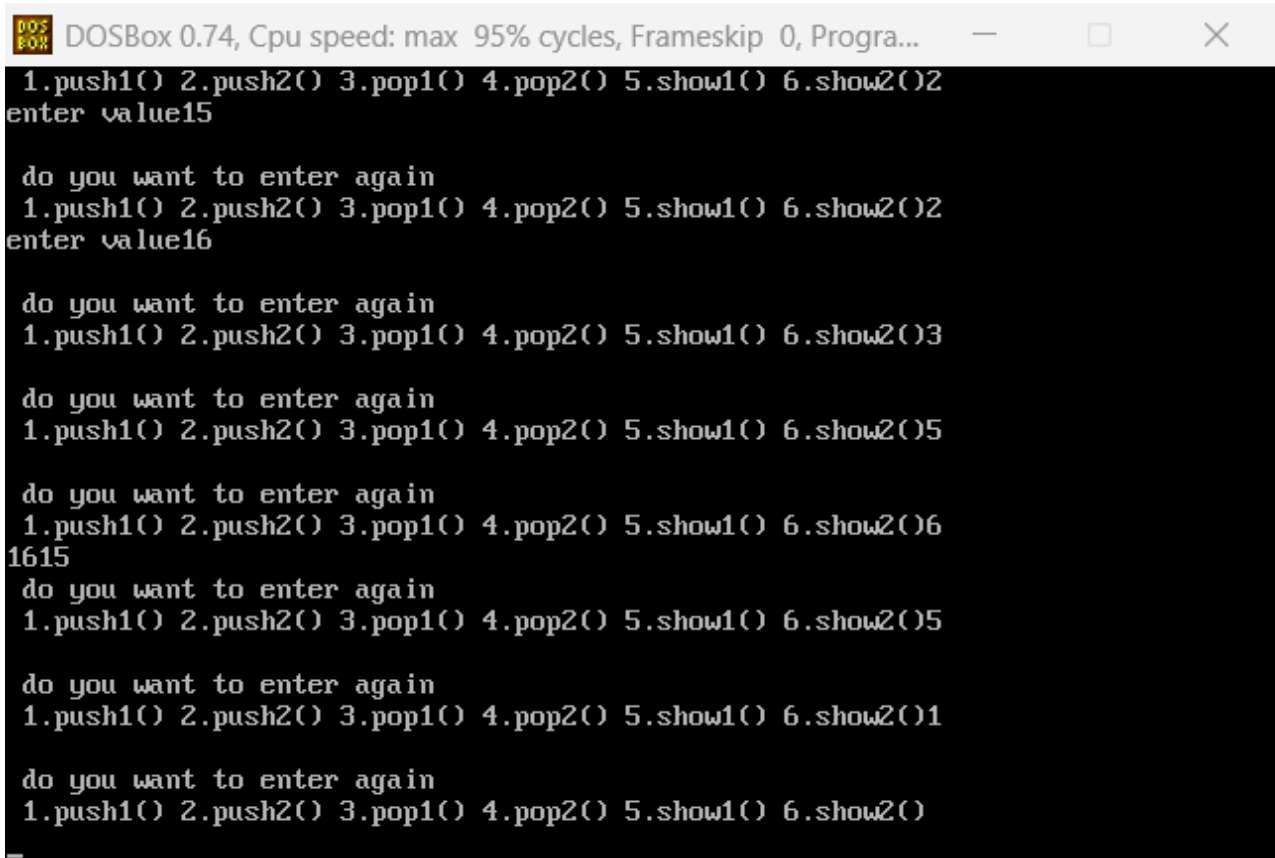
void multiple_stack::show1()
{
    for(int i=top1;i<0;i--)
    {
        cout<<a[i];
    }
}

void multiple_stack::show2()
{
    for(int i=top2;i<=max-1;i++)
    {
        cout<<a[i];
    }
}

void main()
{
    multiple_stack m;
    int choice;
    char c;
    do{
        cout<<"\n 1.push1() 2.push2() 3.pop1() 4.pop2() 5.show1() 6.show2()";
        cin>>choice;
        switch(choice)
        {
            case 1:
                m.push1();
                break;
            case 2:
                m.push2();
                break;
            case 3:
                m.pop1();
                break;
            case 4:
                m.pop2();
                break;
            case 5:
                m.show1();
                break;
            case 6:
                m.show2();
                break;
        }
        cout<<"\n do you want to enter again";
        c=getch();
    }
}

```

```
}  
while (c=='y' || c=='Y');  
getch();  
}
```



DOSBox 0.74, Cpu speed: max 95% cycles, Frameskip 0, Progra...

```
1.push1() 2.push2() 3.pop1() 4.pop2() 5.show1() 6.show2()2  
enter value15  
  
do you want to enter again  
1.push1() 2.push2() 3.pop1() 4.pop2() 5.show1() 6.show2()2  
enter value16  
  
do you want to enter again  
1.push1() 2.push2() 3.pop1() 4.pop2() 5.show1() 6.show2()3  
  
do you want to enter again  
1.push1() 2.push2() 3.pop1() 4.pop2() 5.show1() 6.show2()5  
  
do you want to enter again  
1.push1() 2.push2() 3.pop1() 4.pop2() 5.show1() 6.show2()6  
1615  
do you want to enter again  
1.push1() 2.push2() 3.pop1() 4.pop2() 5.show1() 6.show2()5  
  
do you want to enter again  
1.push1() 2.push2() 3.pop1() 4.pop2() 5.show1() 6.show2()1  
  
do you want to enter again  
1.push1() 2.push2() 3.pop1() 4.pop2() 5.show1() 6.show2()
```